

Education

University of Utah, John and Maria Price College of Engineering – Salt Lake City, UT

BS/MS Biomedical Engineering, Minor in Computer Science

December 2026

Graduate Certificate Deep Learning & AI Robotics

December 2026

University of Utah Presidential Scholar, Daniels Scholar, Tau Beta Pi Member, Dean's List 2021-26, GPA: 3.96

Relevant Coursework: Biomaterials, Biomechanics I & II, Biosignals, BioDesign I & II, Digital Design and Fabrication, Data Management for ML, Deep Learning, Advanced Artificial Intelligence, Simulation of Human Movement

Work Experience

Graduate Research Assistant – Foot & Ankle Mechanics and Morphology (FAMM) Lab

Salt Lake City, UT

May 2024 – Present

- Processed and analyzed patient weight-bearing CT scans using advanced 3D modeling software to quantify bone morphology and joint kinematics.
- Operated, maintained, and trained others on the use of a multi-camera Vicon optical motion capture system to ensure precise and repeatable data collection.
- Managed large datasets of motion-capture and ground reaction force measurements, optimizing them for integration into deep learning pipelines.
- Collaborated with graduate researchers to refine biomechanical analysis workflows, accelerating project timelines.

Lift and Ride Operator - Snowbird Mountain Resort | Salt Lake City, UT

May 2021 – April 2025

- Directed a crew of operators to actively monitor guest activities, enforce safety procedures, and maintain smooth lift operations.
- Inspected, documented, and addressed equipment issues to ensure uninterrupted, safe performance.
- Trained and mentored new hires through hands-on demonstrations, fostering a high-performance team culture.

Technical Skills

Programming Languages: Java, MATLAB, Python (Numpy, Pandas, PyTorch, Sci-Kit Learn), C#

Technical Skills: Materialize Mimics, CAD (Fusion360), Additive Manufacturing, End Milling, Lathe, CNC Machining, Vicon camera system, Musculoskeletal Modeling and Simulation (OpenSim), Deep Learning, Machine Learning, Data Preprocessing, Data Visualization, Relational Databases

Projects

OpenSim Deep Learning Project – University of Utah, FAMM Lab

June 2025 – Present

- To support a growing biomechanics dataset with no standardized pipeline, designed scalable preprocessing workflows for multi-modal data (motion capture, ground reaction forces), cutting data integration time for new trials by eliminating manual formatting steps.
- To generate ground-truth force labels for model training, used OpenSim to simulate each experimental trial and implemented a custom static optimization, producing muscle force estimates across all subjects.
- To enable a robotic controller with live feedback, built and validated deep learning models for older and younger adult populations that predict lower-limb muscle and joint contact forces using ground reaction force data alone.

Automated Nursery Inventory Viability System – University of Utah

January 2026 – Present

- To replace unreliable manual plant grading at commercial nurseries, developed a computer vision framework that automates inventory viability assessment, enabling objective, scalable forecasting across large product volumes.
- To ensure the system would work in real production environments, conducted on-site interviews with commercial nursery staff, translating qualitative grading criteria into precise, implementable technical specifications.

Esophageal SpO2 Monitor Capstone Project – University of Utah

January - November 2024

- As the hardware lead on a multidisciplinary capstone team developing a novel esophageal SpO2 monitor, directed PCB selection, sensor integration, and component sourcing to meet clinical and regulatory feasibility requirements.
- To ensure the device met FDA and ISO 13485 standards ahead of a final design review, owned all design documentation and verification/validation protocols, keeping the team aligned with submission deadlines.

Leadership Experience

Students for the Wasatch: Merchandise coordinator

Fall 2022 – Spring 2023

Boy Scouts: Eagle Scout

2014 - 2018